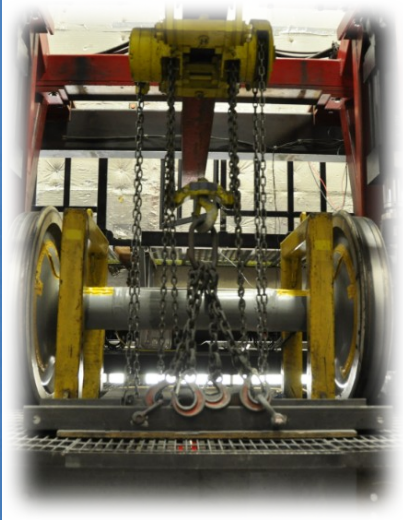




Simulation-Assisted Testing for Railway Technology Validation: Vehicle Component Monitoring and Winter Wheel Damage

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B O S T O N



Overview

- **Engineering Challenge**
Bridging real-world railway operation and laboratory testing
- **Case Study 1**
Vehicle component monitoring system validation
- **Case Study 2**
Winter wheel tread damage investigation
- **Key Takeaways**
Simulation-assisted testing framework

Challenges in Modern Rail Transit

Growing Passenger Demand

- Higher service frequency
- Increased passenger loading
- Reduced maintenance windows

* Harsh Operating Environment

- Snow and ice
- Temperature extremes
- Rain and moisture
- Seasonal operating conditions

⚠ Higher Reliability & Safety Expectations

- Improved service reliability
- Earlier fault detection
- Reduced service disruptions
- Enhanced operational safety

Increasing Demand for Solutions

Advanced Railway Technologies

- Smart monitoring systems
- New sensing technologies
- Advanced materials & components
- Predictive maintenance solutions

Improved Investigation Methodologies

- Root cause investigation
- Failure mechanism studies
- Technology validation
- Controlled failure reproduction

Key Question

How can these technologies and investigation methods be evaluated safely and under representative operating conditions?





Current Evaluation Approaches

Evaluation Criteria

- Representative operating conditions
- Controlled environment
- Repeatable test conditions
- Isolate technology response
- Reproduce the failure condition
- Safe & accessible testing
- Cost & time efficiency

Field testing

Lab testing

Transit Operating Environment (why is reproducing real operation difficult)

Transit Operating Environment

Wheel–rail loading is governed by the combined effects of multiple operating variables.



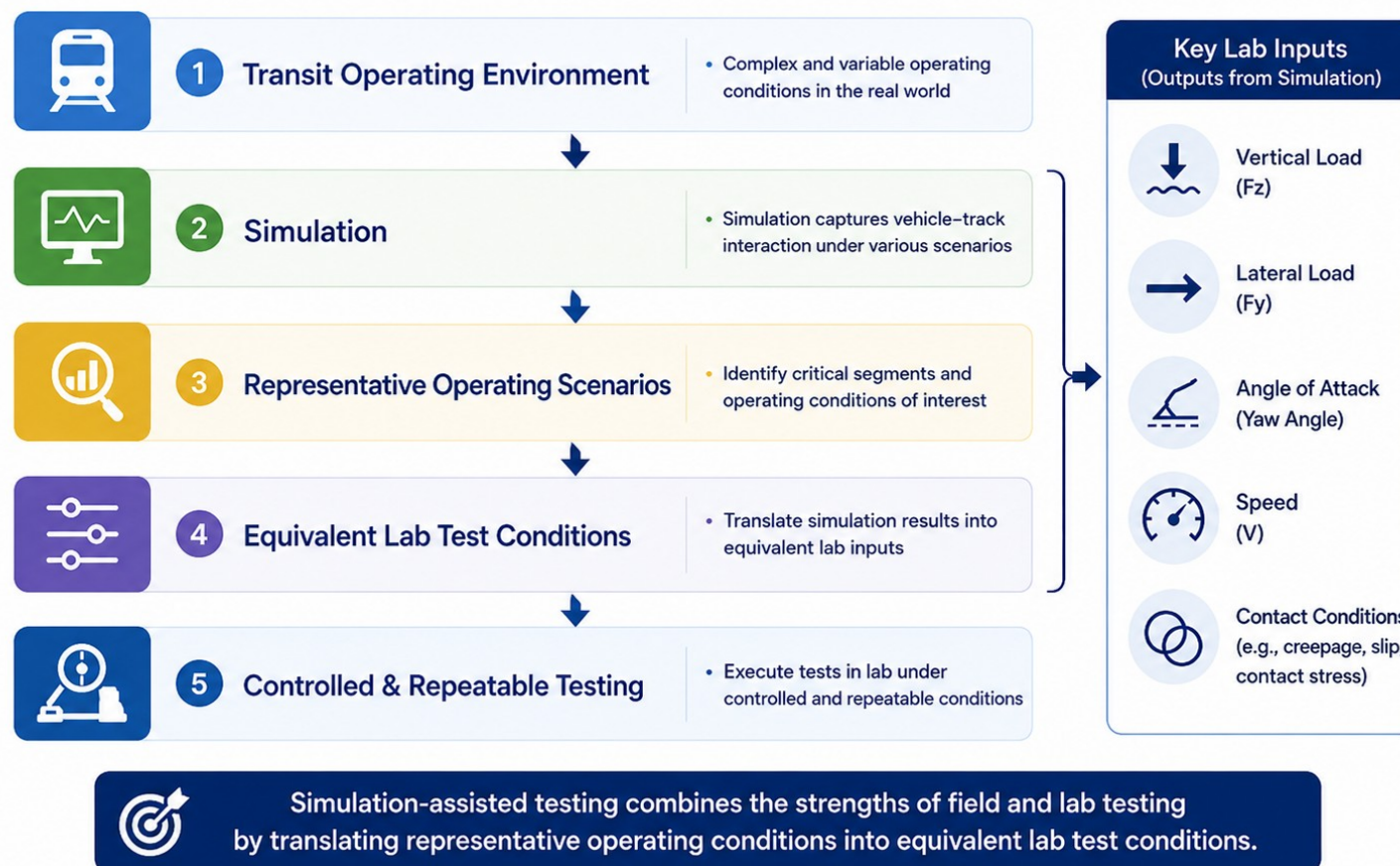
To evaluate railway technologies, laboratory testing should reproduce the key characteristics of the transit operating environment.

(Image generated by AI)

How can lab testing be made representative of real-world operating conditions?

Simulation-Assisted Testing Framework

Bridging real-world operation and controlled lab testing



Wheel Bearing Brake (WBB) Facility at NRC



Key Capabilities

Capability	WBB specification / function
Speed	Up to 150 km/h running speed (upgrade in progress)
Drive power	110 kW (150 hp) electric motor (upgrade in progress)
2 vertical actuators	Up to 356 kN, controlled vertical load to reproduce axle/wheel-rail loading
2 lateral actuators	Up to 173 kN, Independent controlled lateral loading
Yaw / angle of attack	Controlled using a servo hydraulic actuator connected to the test fixture through two longitudinal arms (± 30 milliradian).
Wheelset motion	Allows lateral movement while vertical/lateral loads are applied

<https://nrc.canada.ca/en/research-development/nrc-facilities/wheel-bearing-brake-research-facility>



Case study 1: Validation of a Vehicle Component Monitoring System

- Client objective

Validate a commercial vehicle component vibration monitoring system under representative rail operating conditions before field deployment.

- Client's Initial Test Strategy

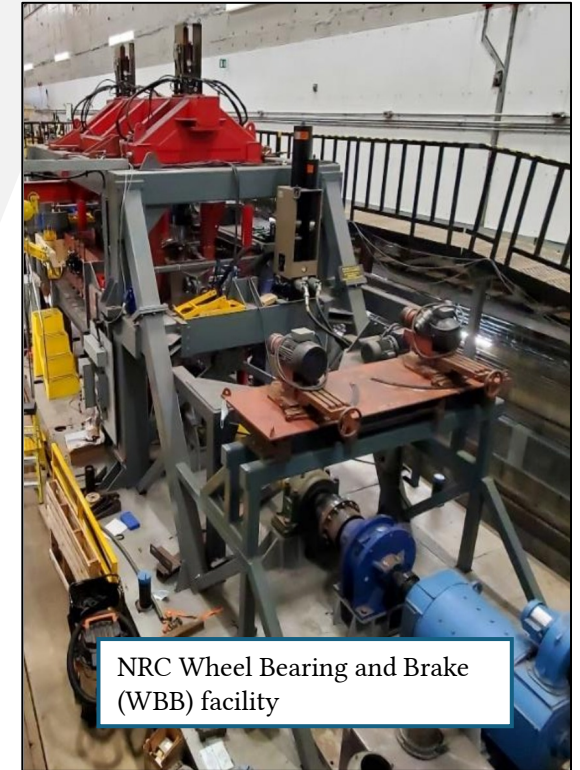
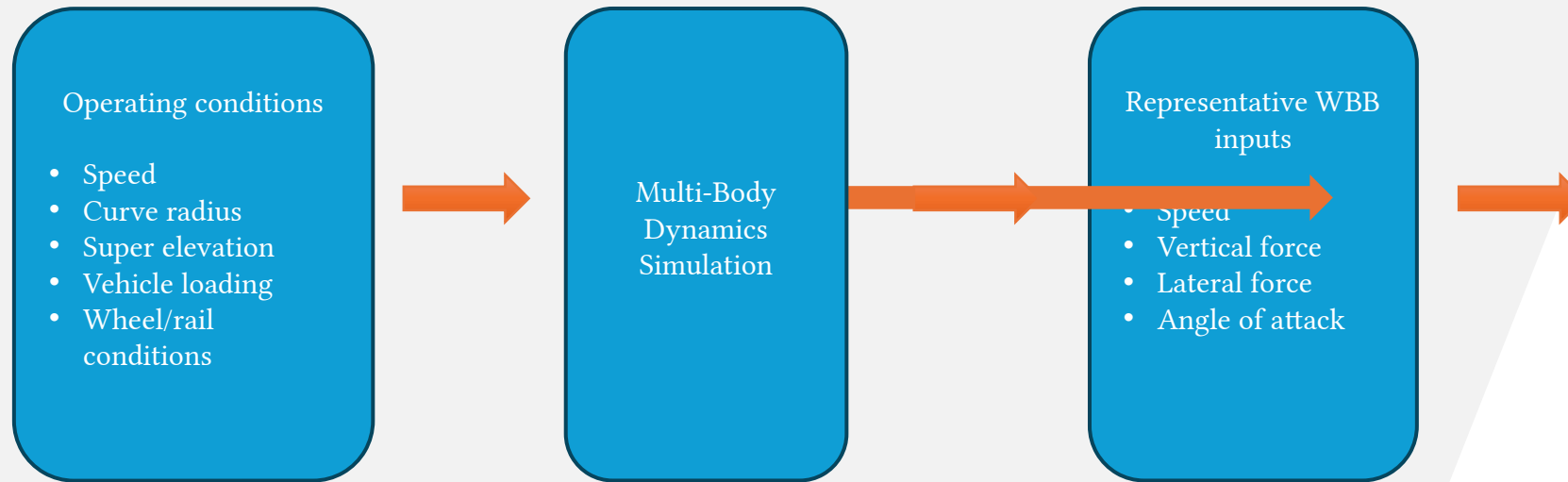
Reproduce the route on the NRC Wheel Bearing and Brake facility.

- Engineering challenge:

How can real operating conditions be translated into representative WBB test inputs without reproducing the entire route?

- Route conditions are not direct WBB inputs.
- Full-route replication is impractical.

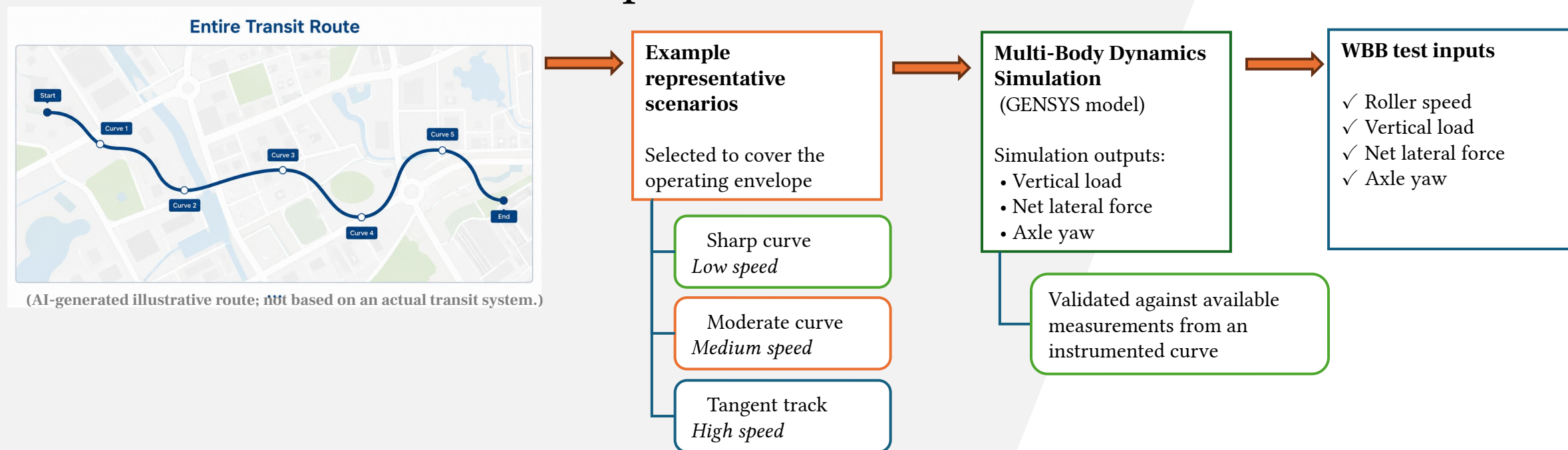
Case study 1: Validation of a Vehicle Component Monitoring System (cont'd)



NRC Wheel Bearing and Brake (WBB) facility

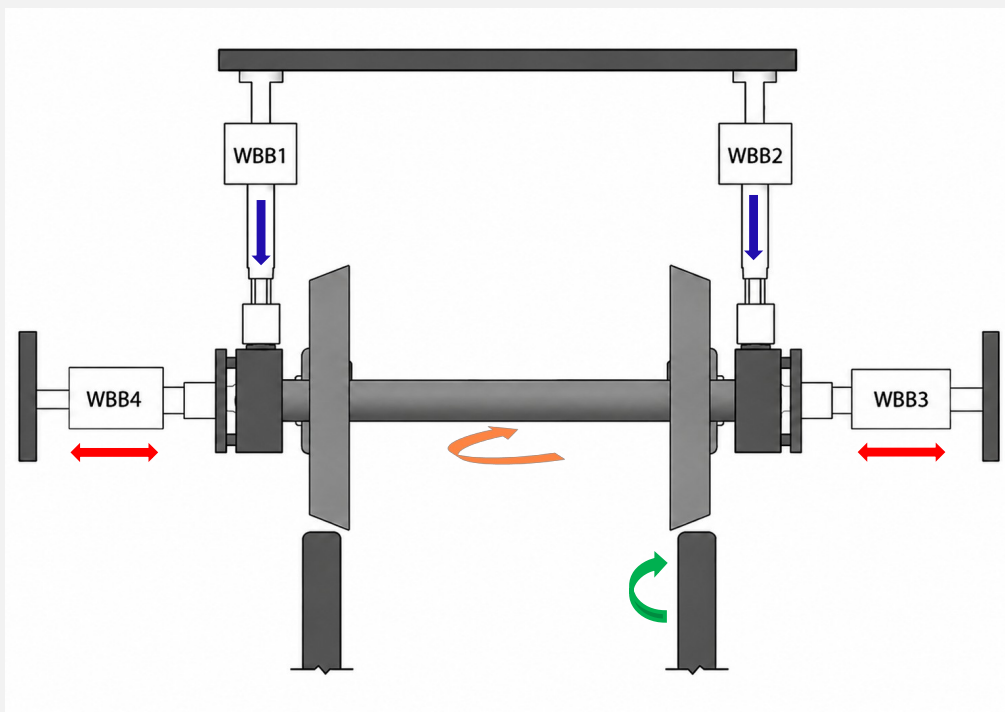
Case study 1: Validation of a Vehicle Component Monitoring System (cont'd)

- From the Entire Route to Representative Lab Test conditions



Selected route scenarios were translated into realistic WBB inputs, avoiding full-route replication.

Case study 1: Validation of a Vehicle Component Monitoring System (cont'd)



- 1 Commercial vibration sensor
- 2 Reference accelerometers
- 3 Lateral displacement
- 4 Vertical displacement

Controlled WBB input from simulation

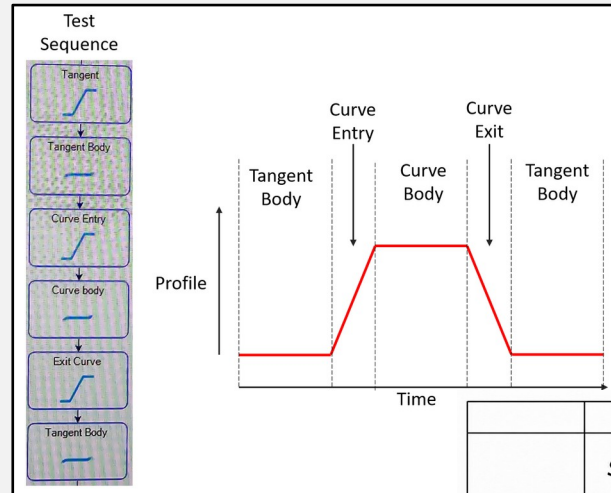
Vertical load Lateral load Axle yaw Roller speed

Simulation-derived loading conditions were applied on the WBB while independent reference measurements were collected to evaluate the commercial monitoring system.



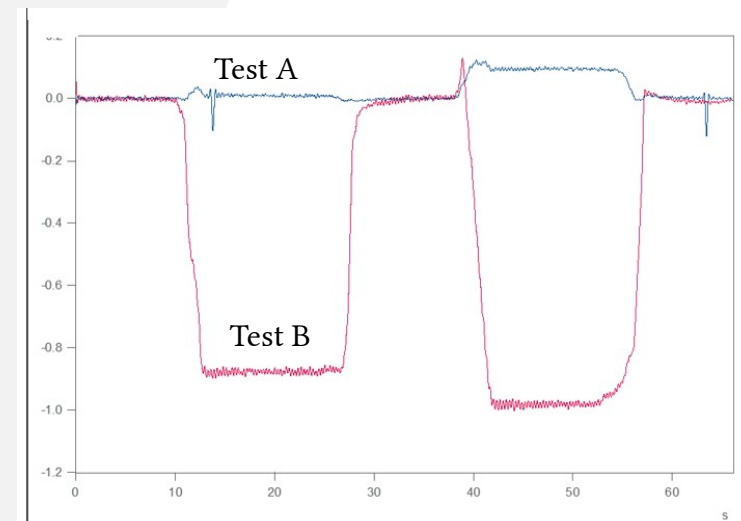
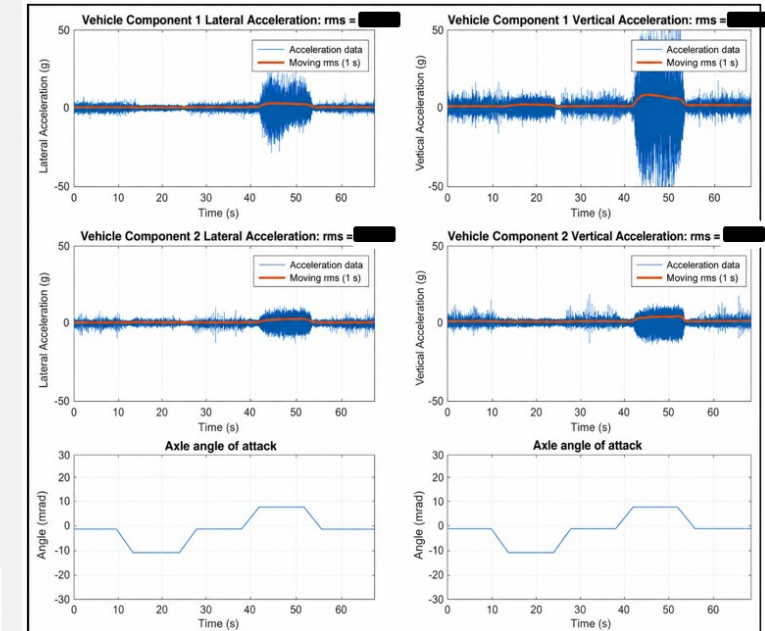
Case study 1: Validation of a Vehicle Component Monitoring System (cont'd)

Example Laboratory Measurements



Trailing Axle: Curving Forces						
	Simulated Curve Radius	speed	High Rail vertical force	Low Rail vertical force	Lateral Force [1]	Axle Yaw (Angle of Attack) [2]
	m	km/h	lb	lb	lb	mrاد
19						
20						
21						

The laboratory platform enables multiple synchronized measurements to support vehicle component condition monitoring investigations.





Case study 2: Investigation of wheel tread damage

- Client problem

Operating during extreme low adhesion (such as braking in winters) showed martensite creation on tread surface, at times metal pickup (MPU) was observed on brake shoes:

- Increased wheel reprofiling
- Reduced braking consistency
- Increased maintenance demand

- Client need

Develop a controlled full-scale test capability to evaluate the client's wheels/mitigation under specified conditions.

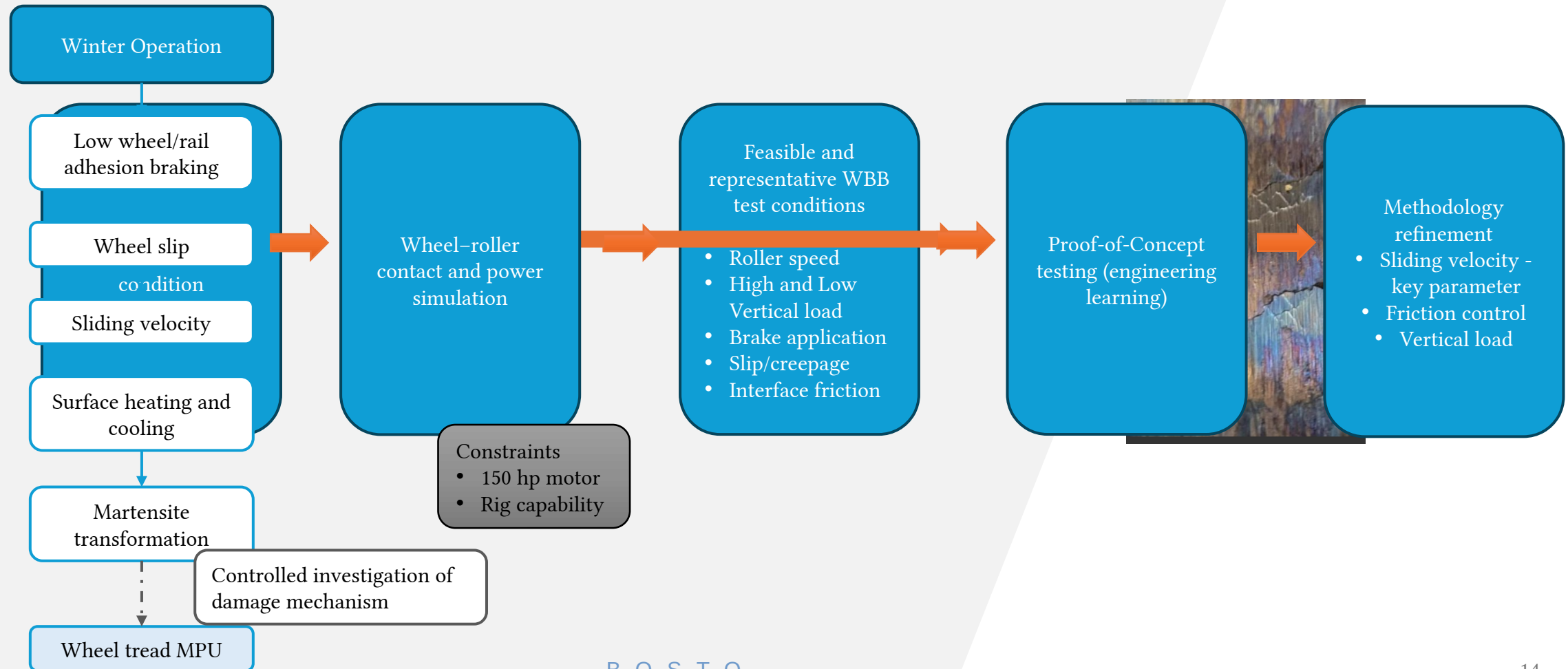
- Our approach

Develop a simulation-assisted full-scale testing methodology to:

- Establish feasible initial test conditions based on power simulation
- Identify and control the parameters required to reproduce the damage mechanism
- Establish a repeatable platform for subsequent wheel testing and mitigation studies.

Case study 2: Investigation of wheel tread damage (cont'd)

- How to create the damage condition on the WBB?



Case study 2: Investigation of wheel tread damage (cont'd)

• Proof-of-Concept test configuration

WBB roller rig:

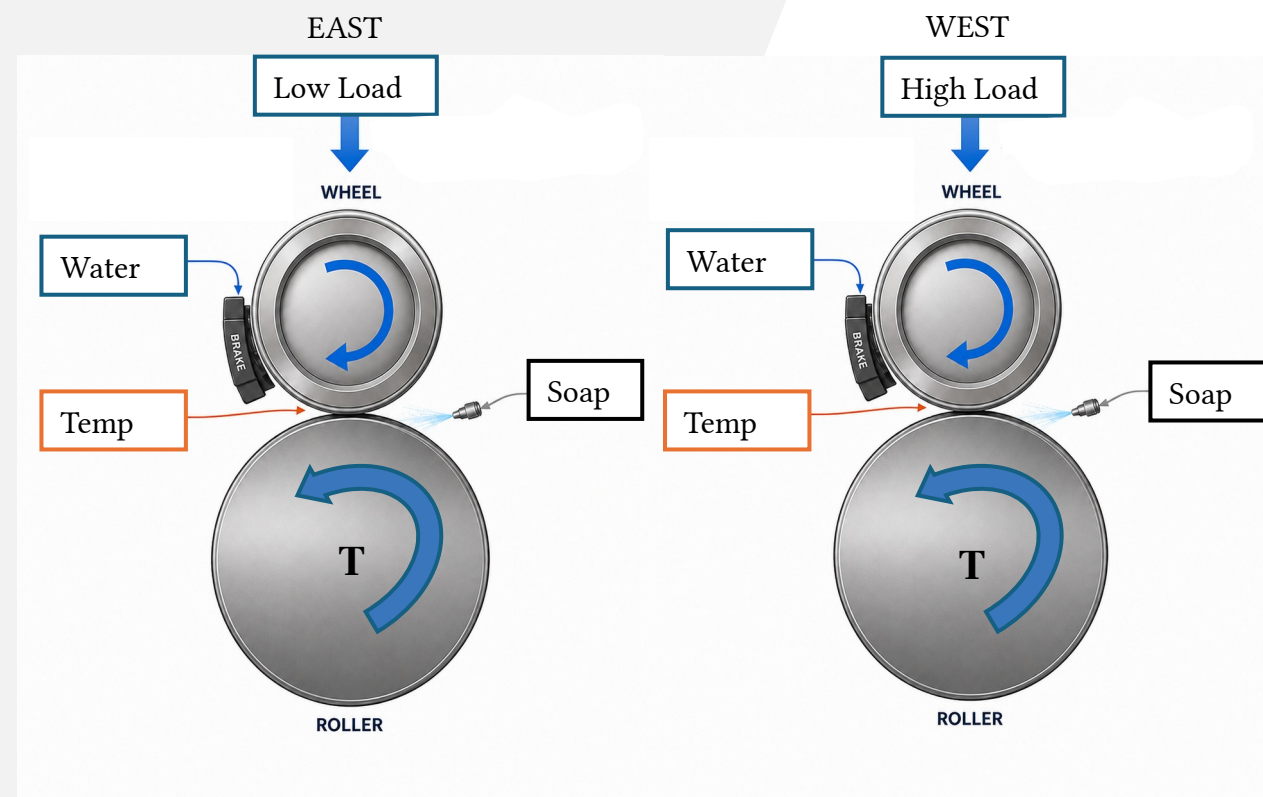
- 150 hp drive motor with 3.12:1 gear reducer
- Fixed roller speed: 45 km/h

Controlled test inputs:

- Roller speed
- Brake force
- Vertical wheel load – High and Low
- Low-adhesion interface conditioning - Soap

Key measurements:

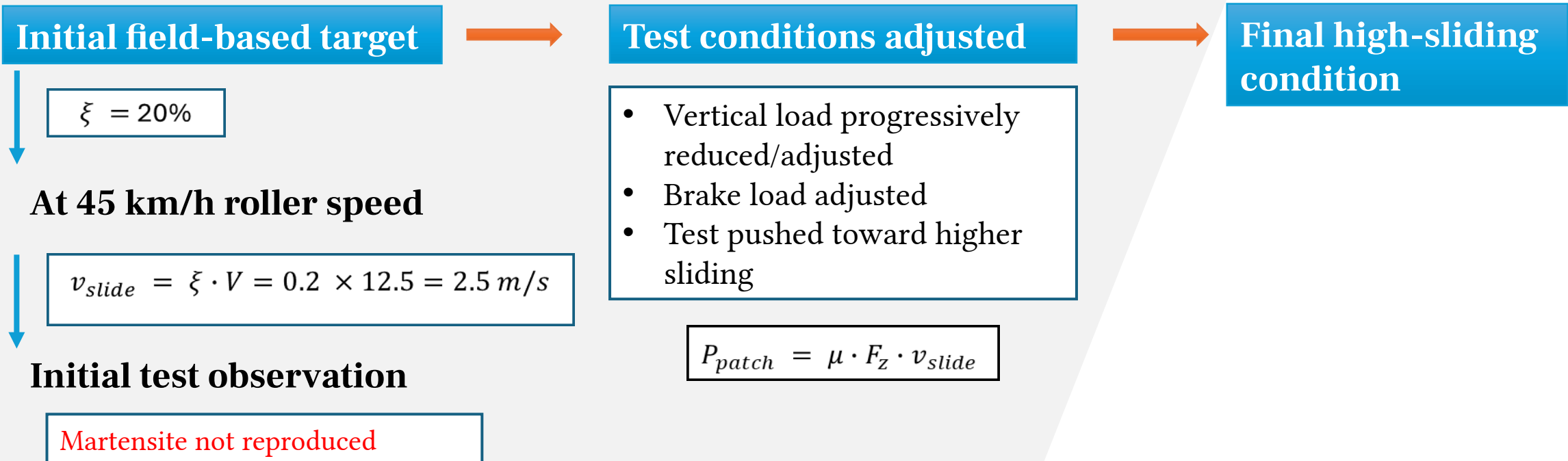
- Wheel surface temperature
- Brake shoe temperature
- Wheel and roller speed
- Brake force and wheel loads



Initial test objective: Reproduce the field-observed wheel damage under low-adhesion braking, using ~20% **creepage** as the initial target condition.

Case study 2: Investigation of wheel tread damage (cont'd)

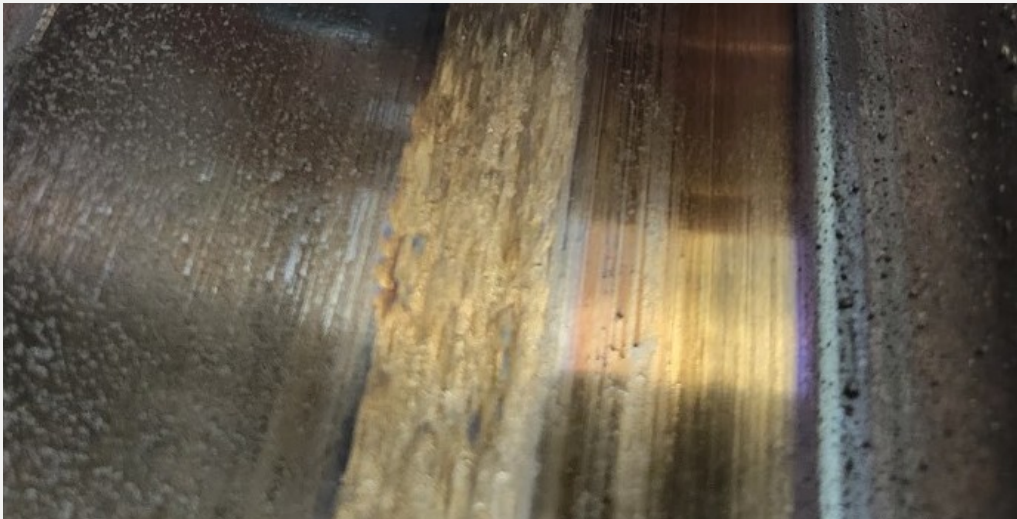
- Was the initial ~20% creepage target sufficient?



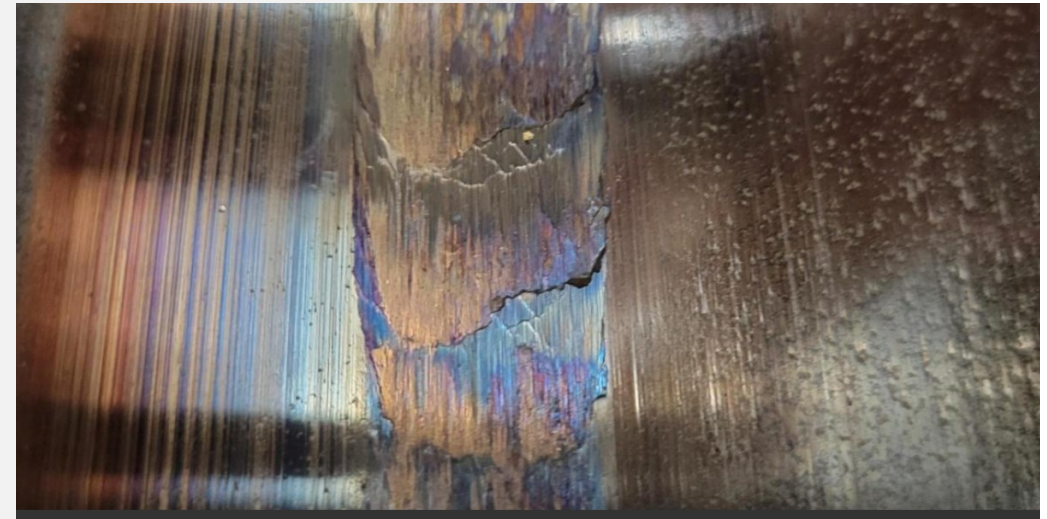
Case study 2: Investigation of wheel tread damage (cont'd)

- Martensite observed on both wheels

East Wheel (Low load)



West Wheel (High load)



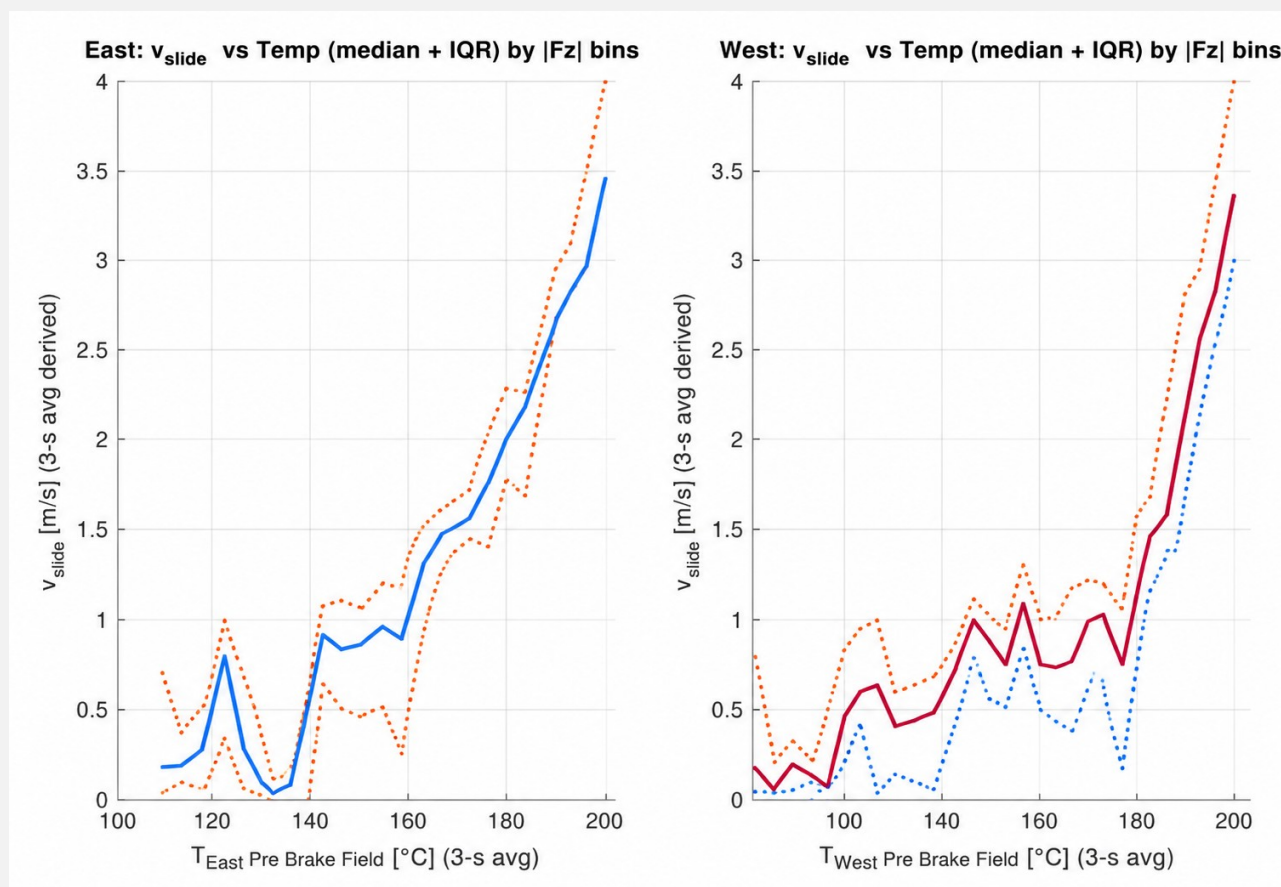
Hardness result

- Before: ~360 HB
- After: ~600 HB
- Wheel damage matched client field observations



Case study 2: Investigation of wheel tread damage (cont'd)

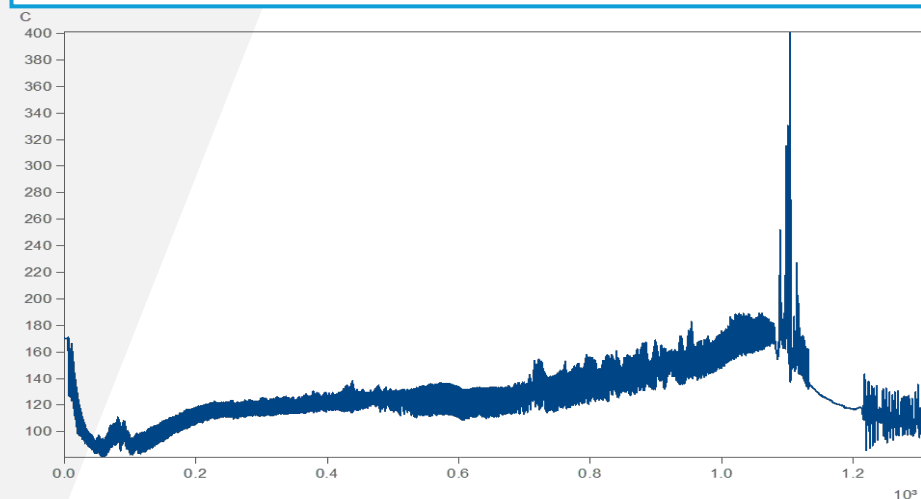
- What was the successful condition?



Note: Measured temperature does not represent instantaneous contact flash temperature.

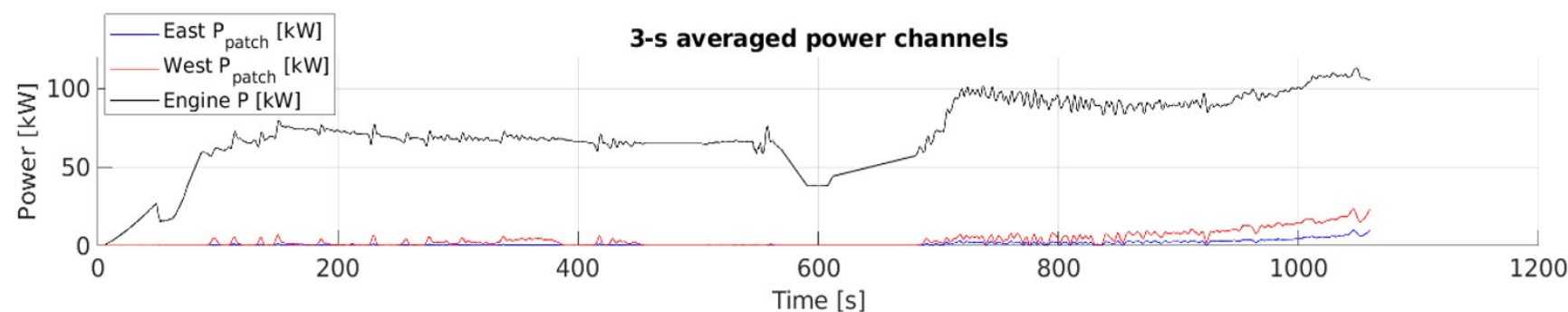
Post-test data analysis:

- Measured wheel temperature increased substantially with sliding velocity.
- The highest sliding velocities (~ 4 m/s) coincided with the most severe measured thermal response.
- **High sliding velocity was associated with the severe thermal condition that produced martensite.**



Case study 2: Investigation of wheel tread damage (cont'd)

- How was the sliding condition achieved within the motor-power limit?

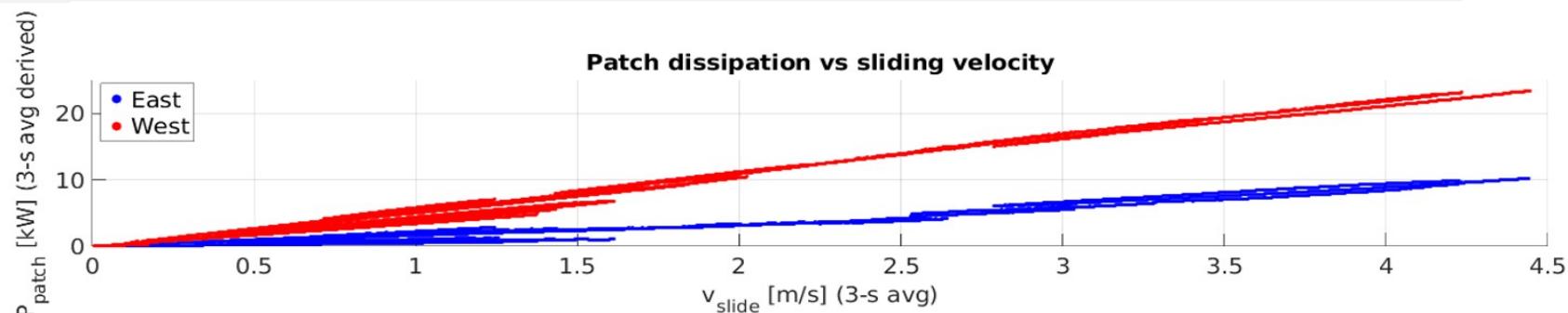


At ~4 m/s:

West patch: ~20–25 kW

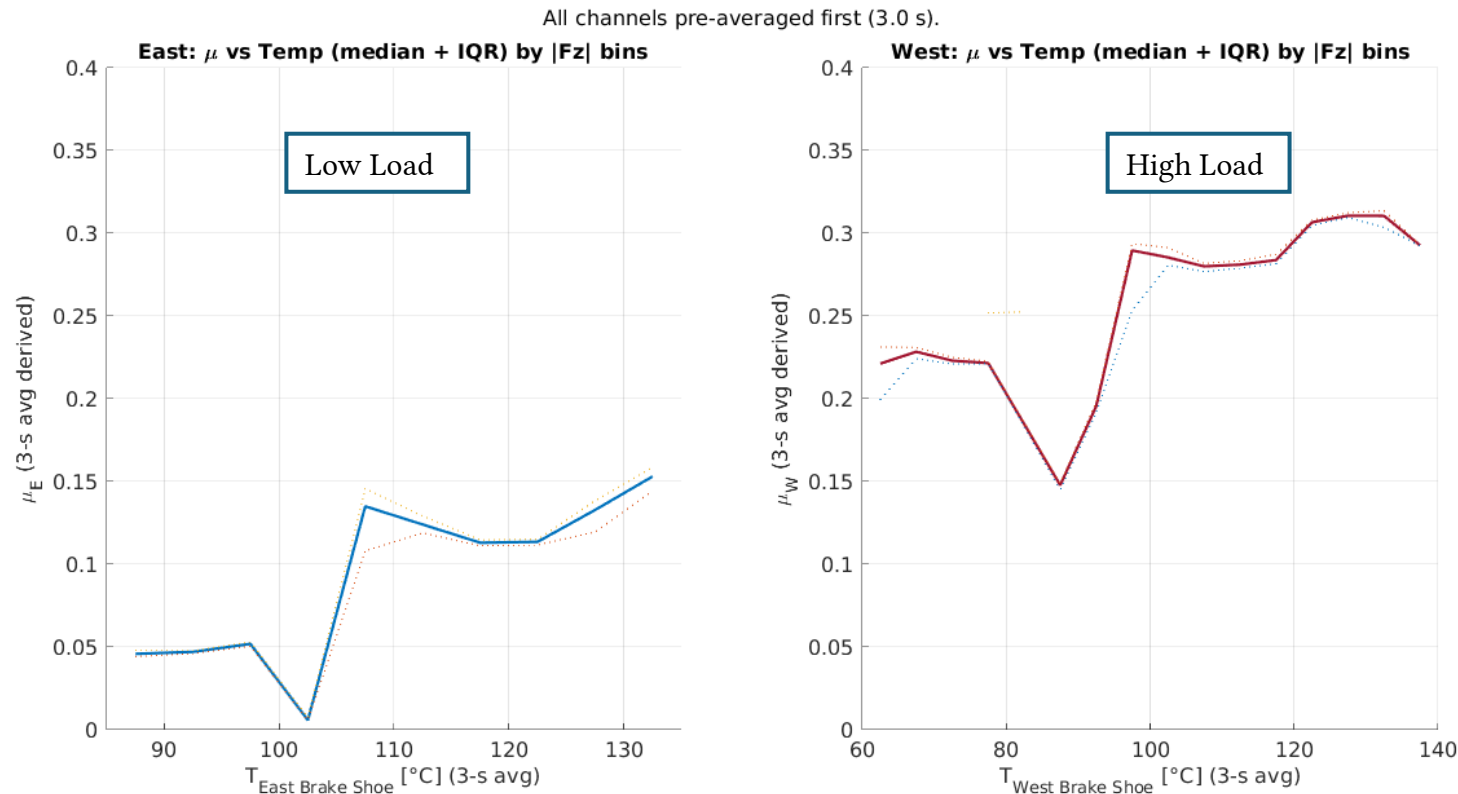
East patch: ~9–11 kW

Motor power: approaching the available drive-power limit



Case study 2: Investigation of wheel tread damage (cont'd)

- Why contact patch power was different between East and West?

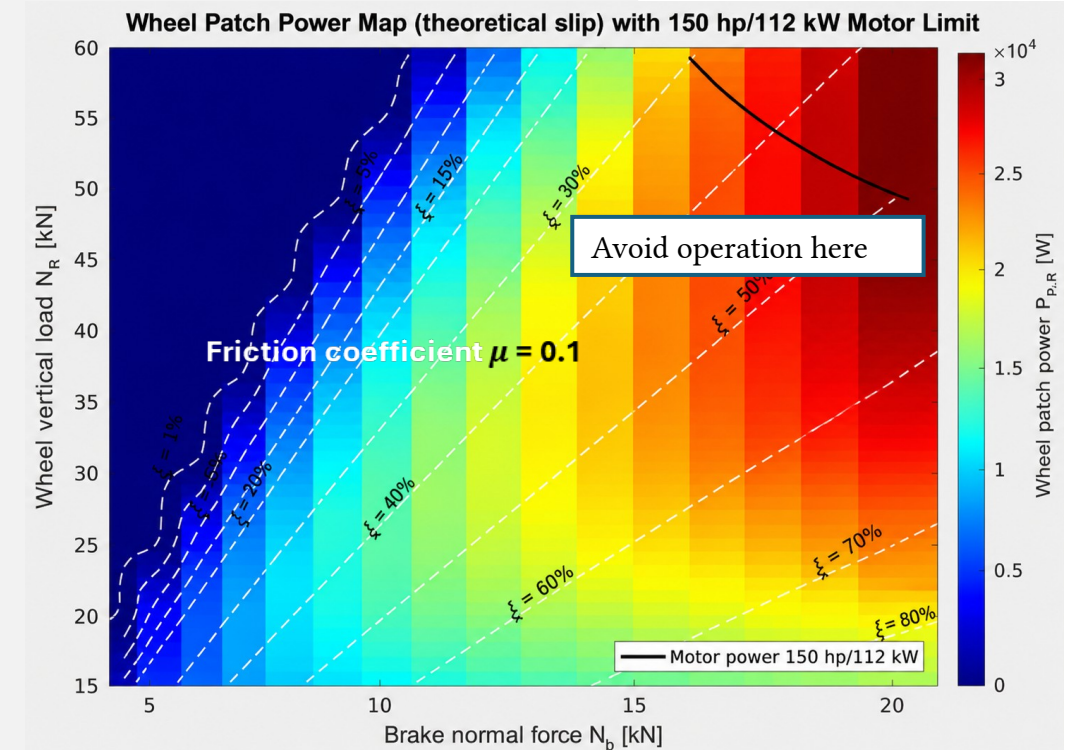
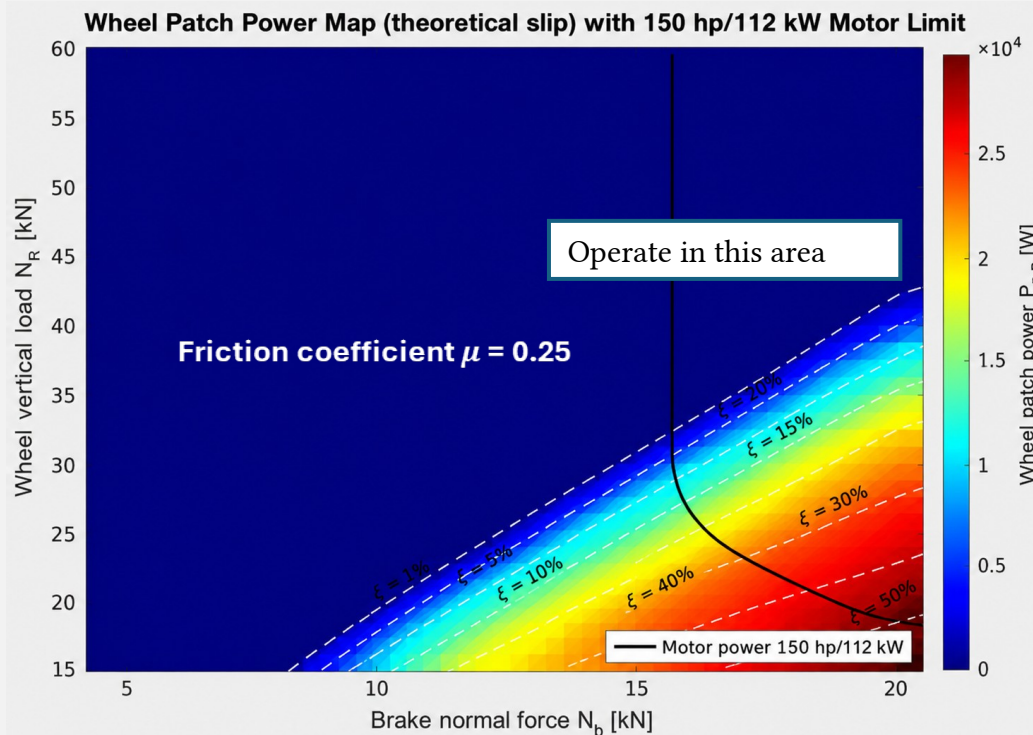


Brake from high Load

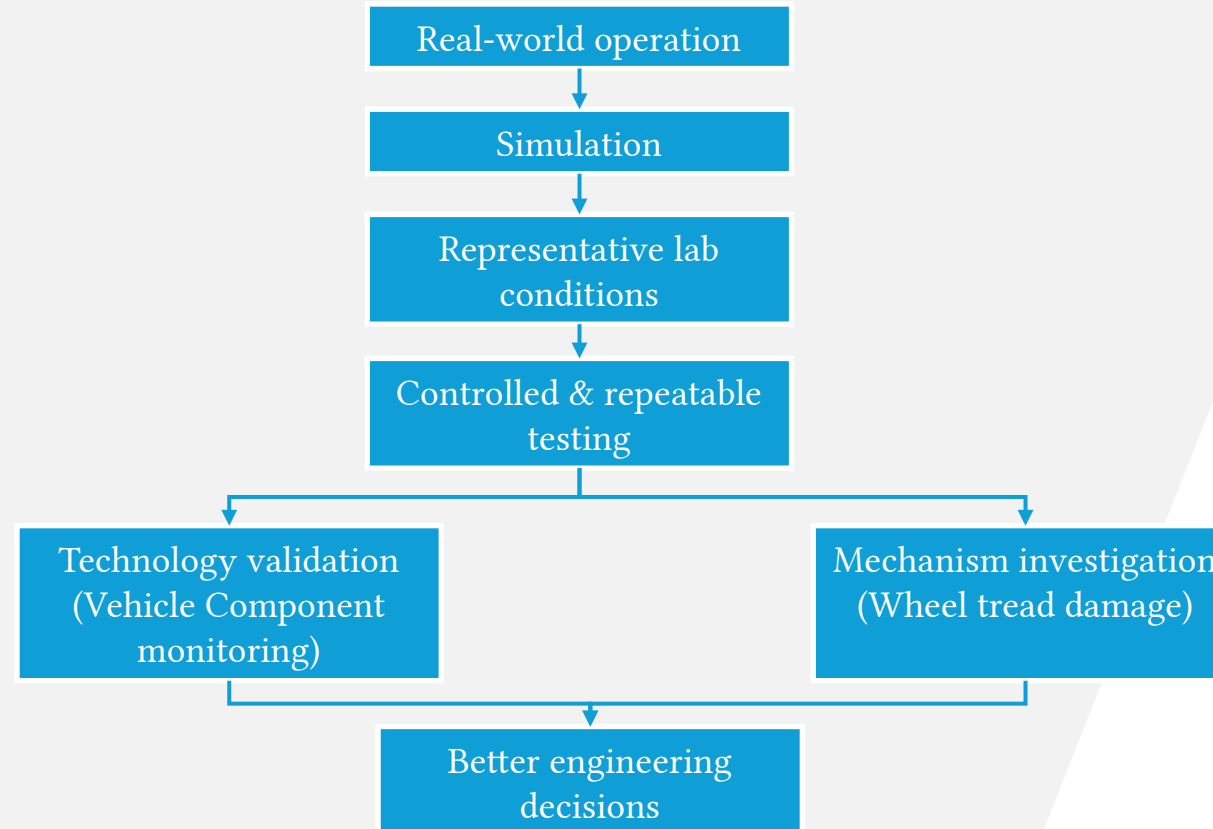


Case study 2: What does this mean for transit operations

- Engineering insight: Why friction control matters



Simulation-Assisted Testing: Key Takeaways



- Simulation translates real-world operating conditions into representative test inputs.
- Controlled full-scale lab testing enables repeatable validation and direct investigation of engineering mechanisms.
- Combining simulation with full-scale testing reduces development risk and increases confidence before field deployment.



Thank You

- Authors:
 - Shuwen Gao (Test plan and testing),
 - Kevin Mackie (WBB planning and operation),
 - Saeed H-Nia (Simulation),
 - Elton Toma (Concept and oversight).
- Contributors:
 - Jean-François Lemieux (Project manager),
 - Instrumentation team at AST.
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QUESTIONS?